

Area Types:

In OSPF (Open Shortest Path First), **areas** are used to segment a large network into smaller, more manageable sections. This segmentation helps reduce routing overhead and keeps the network more scalable. Here's a breakdown of the different OSPF area types:

1. **Backbone Area (Area 0)**
2. **Standard Area.**
3. **Stub Area.**
4. **Totally Stubby Area.**
5. **NSSA.**
6. **Totally NSSA.**

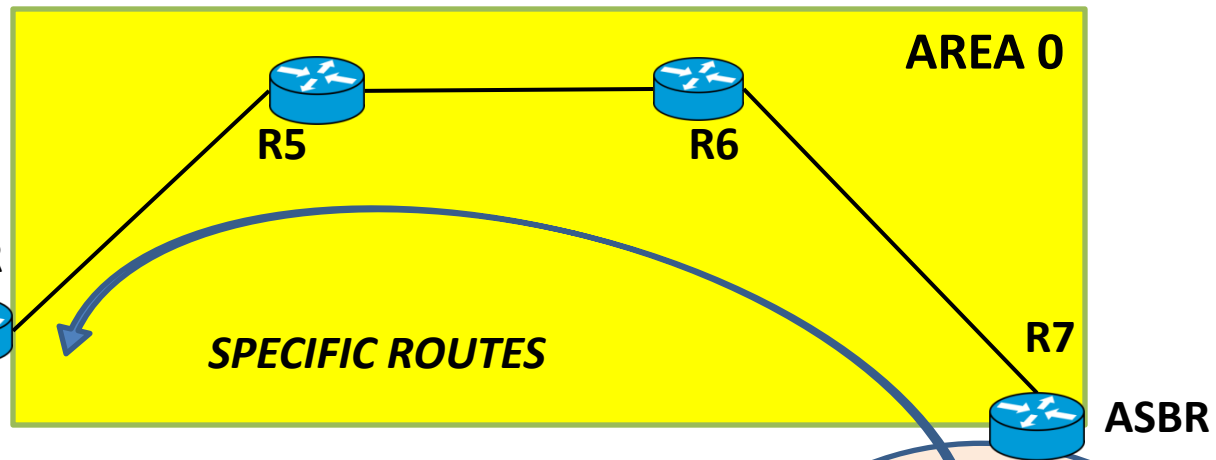
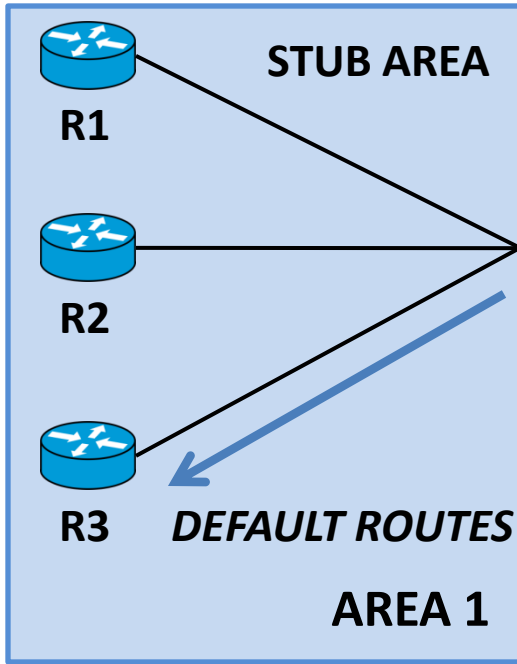
Stub Area:-

Blocks External Routes (Type 5 LSAs): A stub area does not allow Type 5 LSAs from outside the OSPF autonomous system, such as from BGP). This helps to reduce the size of the routing table and the complexity of routing within the area.

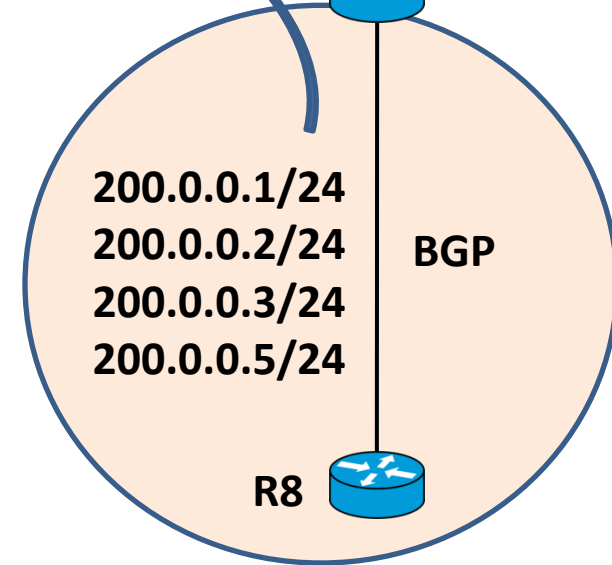
Allows Inter-Area Routes (Type 3 LSAs): A stub area allows Type 3 LSAs, which advertise summary routes from other areas within the same OSPF autonomous system.

Receives a Default Route (Type 3 LSA with 0.0.0.0/0): To handle traffic destined for external networks, a stub area receives a default route (0.0.0.0/0) from the **Area Border Router (ABR)**.

STUB AREA:



Routing Table:
 $E1/E2 \rightarrow o^*IA$



*All the external routes replaced with
Single default route...*

Areas that cannot be stubs:

1. Backbone area 0
2. Virtually connected Non Backbone areas
3. The area with ASBR

Configuration of Stub Area:

Router1(config)#router ospf 1

Router1(config-router)#area 1 stub

Router2(config)#router ospf 1

Router2(config-router)#area 1 stub

Router3(config)#router ospf 1

Router3(config-router)#area 1 stub

Areas that cannot be stubs:

1. Backbone area 0
2. Virtually connected Non Backbone areas
3. The area with ASBR

Configuration of Stub Area:

Router1(config)#router ospf 1

Router1(config-router)#area 1 stub

**Configuration must be done on all the routers included the ABR Router.*

Note* Area Border Router will Maintain the routing table for external areas

Totally Stub Area:-

A **Totally Stubby Area** (TSA) is a type of OSPF (Open Shortest Path First) area designed to further reduce the routing table size and complexity compared to a regular Stub Area. It is a Cisco-proprietary feature, which means it's specific to Cisco devices.

Key Characteristics of a Totally Stubby Area:

Blocks Type 3 LSAs:

In this, Type 3 LSAs (Summary LSAs) are not allowed. Type 3 LSAs carry inter-area routes from other OSPF areas.

Blocks Type 5 LSAs:

Type 5 LSAs (External LSAs) are also blocked in a Totally Stubby Area. Type 5 LSAs carry routes from outside the OSPF Autonomous System (e.g., from BGP or other external sources).

Totally Stubby Area:-

Default Route:

The Area Border Router (ABR) injects a default route (0.0.0.0/0) into the Totally Stubby Area. This default route is used by routers in the area to send traffic to destinations outside the area.

Configuration:

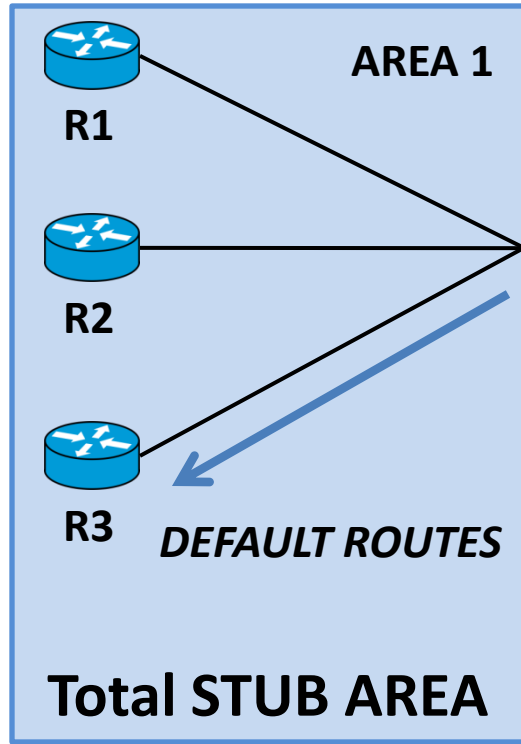
The configuration of a Totally Stubby Area involves setting up the area type on the ABR and the internal routers within the area.

Routing Codes Will not shown in table:

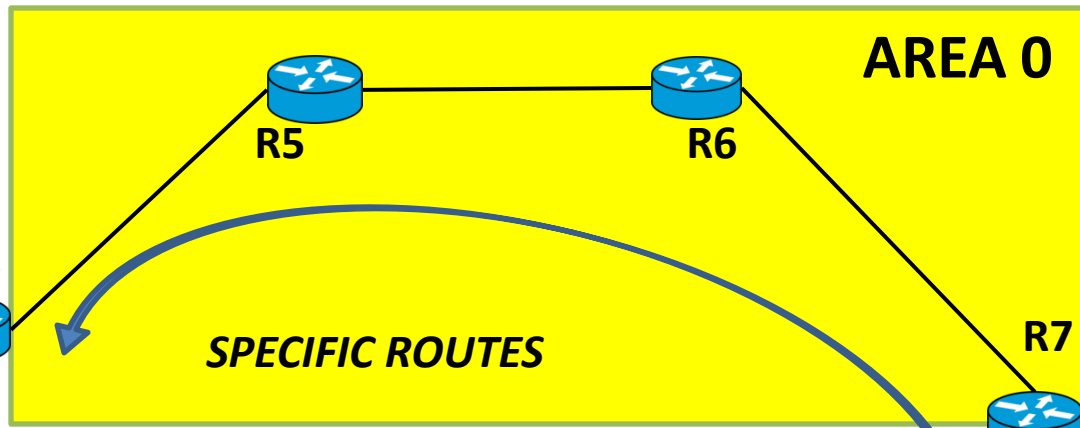
Block: O IA and E1/E2

Note* Type 3 LSAs are not converted to default routes; they are blocked.

Total Stub Area:

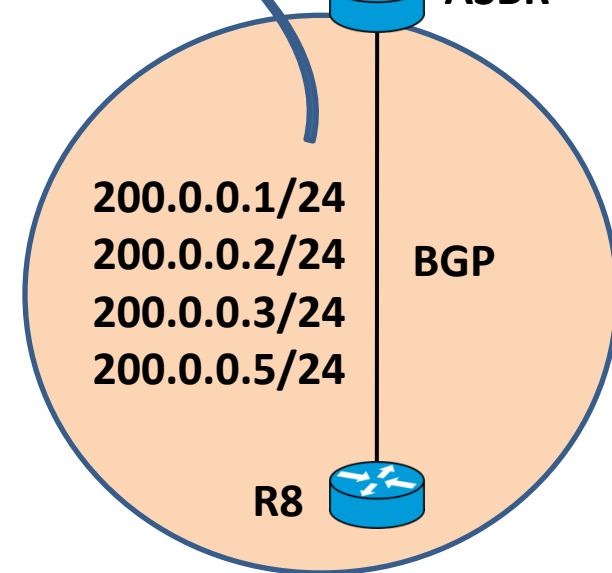


*All the external routes replaced with
Single default route...*



*Block LSA 3 (Inter area routes)
Block LSA 5*

Routing Table:
E1/E2 → o*IA



Totally Stubby Area:-

Configuration Steps:

1. Configure the ABR:

```
Router(Config)#Router router ospf 1 network 10.0.0.0 0.0.0.255 area 0
```

```
Router(config-router)# network 192.168.2.0 0.0.0.255 area 2
```

```
Router(config-router)# area 2 stub no-summary
```

Note* Add ***no-summary*** with Stub Command on the ABR Router Config. Else everything is same as the internal router config.

2. Configure Internal Routers in the Totally Stubby Area:

```
Router(Config)#router ospf 1
```

```
Router(config-router)#network 192.168.2.0 0.0.0.255 area 2
```

```
Router(config-router)#area 2 stub
```

Not So Stubby Area:

A **Not-So-Stubby Area (NSSA)** is a type of OSPF (Open Shortest Path First) area that allows for the injection of external routes while still maintaining some of the benefits of a stub area. This type of area is particularly useful when you need to connect an OSPF network to an external network (e.g., a network running BGP) but want to limit the amount of external routing information propagated into the OSPF domain.

Key Characteristics of an NSSA:

- **Type 7 LSAs.**
- **Allows External Routes.**
- **Blocks Type 5 LSAs.**
- **Default Route.**

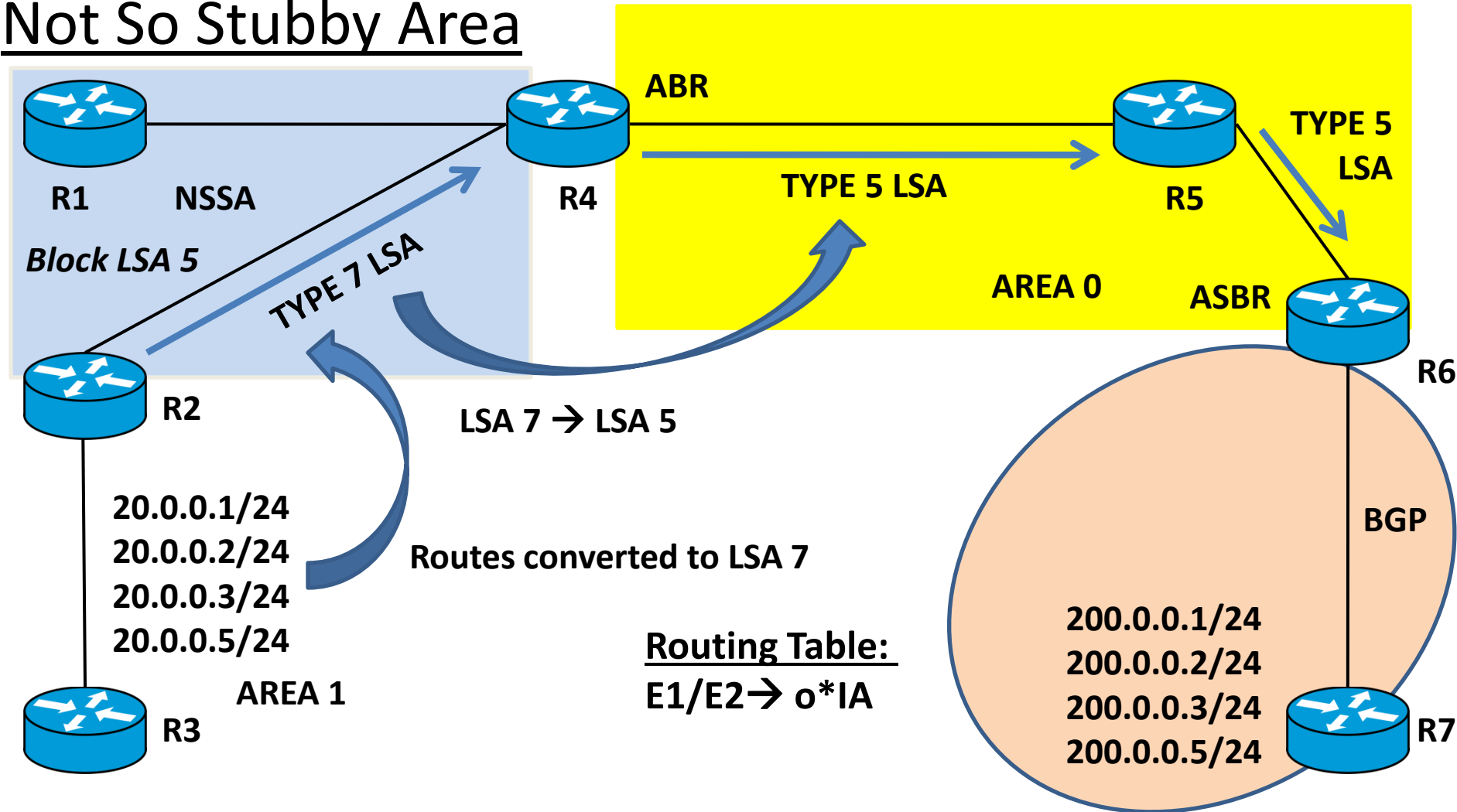
Not So Stubby Area:

Purpose: NSSAs use Type 7 LSAs to carry external routing information. Type 7 LSAs are similar to Type 5 LSAs (used in standard OSPF areas for external routes) but are specific to NSSAs.

- Allow Type 3 LSA
- Blocks Type 5 LSA

Conversion: Type 7 LSAs are eventually converted to Type 5 LSAs when they leave the NSSA and enter other areas of the OSPF network. This conversion happens at the ABR (Area Border Router) that connects the NSSA to other OSPF areas.

Not So Stubby Area



Not So Stubby Area:

Allows External Routes:

External Routing: Unlike stub areas, NSSAs allow for the introduction of external routes (from other routing protocols) into the OSPF domain using Type 7 LSAs, injected by an ASBR within the NSSA.

Blocks Type 5 LSAs:

Internal Routing: NSSAs do not allow Type 5 LSAs, which are used for external routes in standard OSPF areas. This helps to prevent the flooding of external routing information into the NSSA.

Default Route:

Default Route Injection: NSSAs can also receive a default route (0.0.0.0/0) from the ABR if configured to do so. This provides a path for traffic destined for destinations outside the NSSA when no specific route is available.

Not So Stubby Area Configuration in OSPF:-

Configuration Steps:

1. Configure the ABR:

```
Router(Config)#Router router ospf 1 network 10.0.0.0 0.0.0.255 area 0
```

```
Router(config-router)# network 192.168.2.0 0.0.0.255 area 2
```

```
Router(config-router)# area 3 NSSA
```

2. Configure the ASBR

```
Router(Config)#router ospf 1
```

```
Router(config-router)#network 192.168.2.0 0.0.0.255 area 2
```

```
Router(config-router)# ip ospf 1 area 2
```

Totally Not So Stubby Area:

In a **Totally Not-So-Stubby Area (Totally NSSA)**, routers are able to communicate with destinations outside the area because the **ABR (Area Border Router)** explicitly injects a **default route (0.0.0.0/0)** into the area. the OSPF areas without needing specific routing in This default route allows routers inside the Totally NSSA to forward traffic to external networks or formation from **Type 3** or **Type 5 LSAs**, which are blocked.

Here's how communication works in a Totally NSSA:

- 1. Type 3 LSAs Blocked:** Type 3 LSAs, used to propagate inter-area routes from other areas, are **not allowed** into the Totally NSSA.
- 2. Type 5 LSAs Blocked:** Type 5 LSAs, also **blocked** from entering the area.

Totally Not So Stubby Area:

3. Here's how communication works in a Totally NSSA:

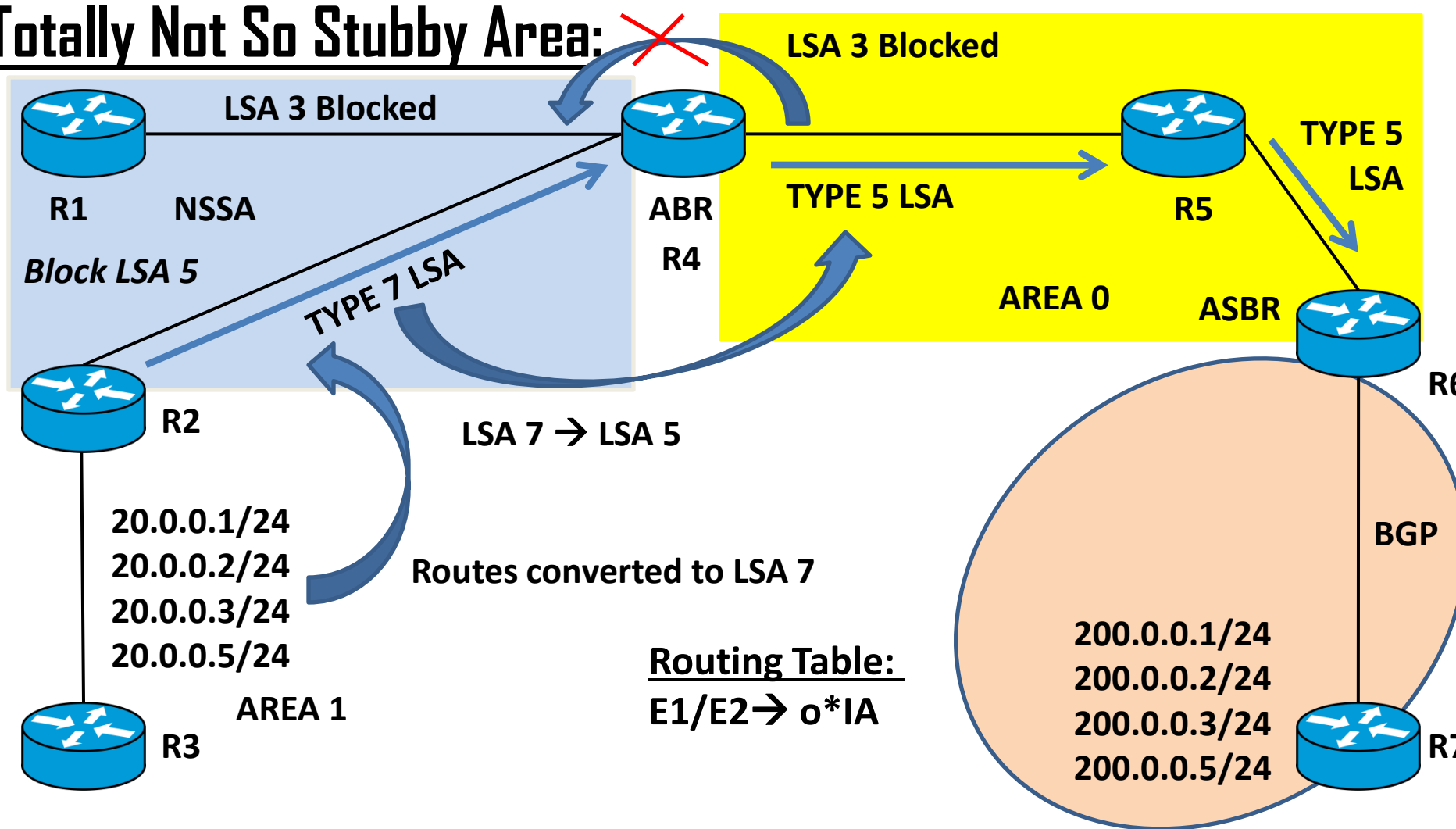
Type 7 LSAs for External Routes:

- If there is an **ASBR (Autonomous System Boundary Router)** inside the Totally NSSA, it can inject external routes using **Type 7 LSAs**. These Type 7 LSAs carry external routes from outside OSPF (e.g., BGP) into the NSSA.
- When these Type 7 LSAs reach the ABR, they are converted into **Type 5 LSAs** to be propagated into other areas.

4. ABR Injects a Default Route (0.0.0.0/0)

- Instead of receiving detailed routing information (like Type 3 or Type 5 LSAs), routers in the Totally NSSA receive a **default route (0.0.0.0/0)** from the ABR.

Totally Not So Stubby Area:



Not So Stubby Area Configuration in OSPF:-

Configuration Steps:

1. Configure the ABR:

```
Router(Config)#Router router ospf 1
```

```
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

```
Router(config-router)# network 192.168.2.0 0.0.0.255 area 2
```

```
Router(config-router)# area 2 nssa no-summary
```

2. Configure the ASBR

```
Router(Config)#router ospf 1
```

```
Router(config-router)#network 192.168.2.0 0.0.0.255 area 2
```

```
Router(config-router)# ip ospf 1 area 2
```